

8. Brand analysis: analyze the conversation around a particular brand for given dataset

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from wordcloud import WordCloud
import nltk
from nltk.tokenize import word_tokenize
from nltk.corpus import stopwords
from nltk.sentiment import SentimentIntensityAnalyzer
from textblob import TextBlob
import re
from collections import Counter
```

```
df = pd.read_csv('location.csv')
df.head()
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtag
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning
2	2	2	Just finished an amazing workout! 💪 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout
3	3	3	Excited about the upcoming weekend	Positive	2023-01-15	AdventureX	Facebook	#Travel #Adventure

Next steps:

[Generate code with df](#)
[View recommended plots](#)
[New interactive sheet](#)

```
nltk.download('wordnet')
nltk.download('stopwords')
nltk.download('punkt')
nltk.download('punkt_tab')
nltk.download('vader_lexicon')
```

```
➦ [nltk_data] Downloading package wordnet to /root/nltk_data...  
[nltk_data] Downloading package stopwords to /root/nltk_data...  
[nltk_data]   Unzipping corpora/stopwords.zip.  
[nltk_data] Downloading package punkt to /root/nltk_data...  
[nltk_data]   Unzipping tokenizers/punkt.zip.  
[nltk_data] Downloading package punkt_tab to /root/nltk_data...  
[nltk_data]   Unzipping tokenizers/punkt_tab.zip.  
[nltk_data] Downloading package vader_lexicon to /root/nltk_data...  
True
```

```
stop_words = set(stopwords.words('english'))  
sia = SentimentIntensityAnalyzer()
```

```
def pre_process(text):  
    tokens = word_tokenize(text.lower())  
    filtered_tokens = [token for token in tokens if token.isalnum() and token not in stop  
    return filtered_tokens
```

```
df['tokens'] = df['Text'].apply(pre_process)  
df['Sentiment_new'] = df['Text'].apply(lambda x: TextBlob(x).sentiment.polarity)  
df.head()
```



	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtag
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1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning
2	2	2	Just finished an amazing workout! 💪 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout
3	3	3	Excited about the upcoming weekend getaway! ...	Positive	2023-01-15 18:20:00	AdventureX	Facebook	#Travel #Adventure
4	4	4	Trying out a new recipe for dinner tonight. ...	Neutral	2023-01-15 19:55:00	ChefCook	Instagram	#Cooking #Food

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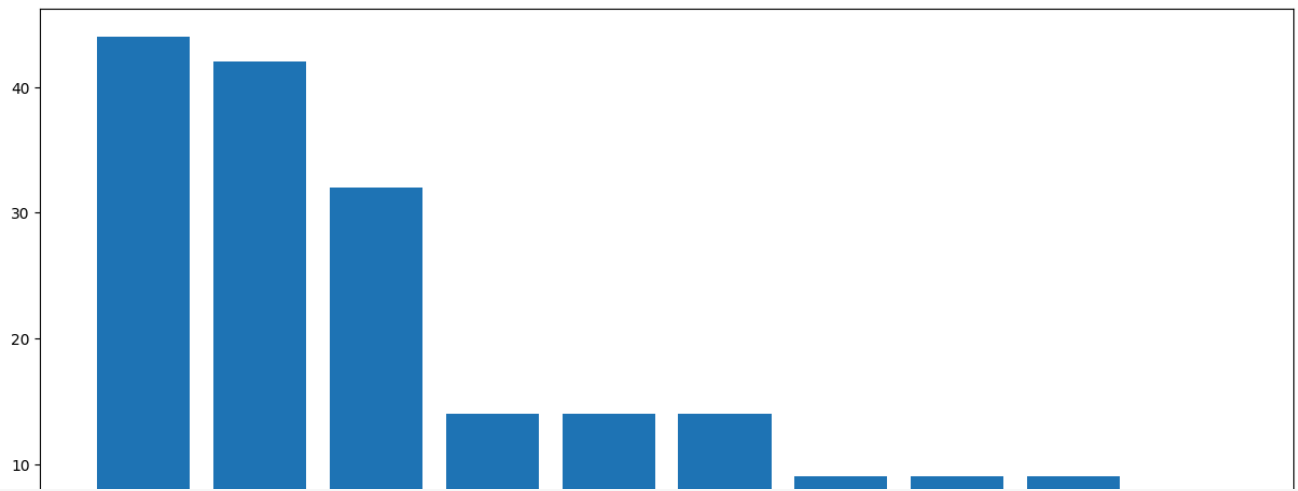
```
all_text = ' '.join(df['Text'])
wordcloud = WordCloud(width=800, height=500, background_color='white').generate(all_text)

plt.figure(figsize=(10, 6))
plt.imshow(wordcloud, interpolation='bilinear')
plt.title("WordCloud of Brand Mentions")
plt.axis('off')
plt.show()
```



➡ Average sentiment of brand mentions: 0.10

4/5



```
hashtags = []
for text in df['Hashtags'].dropna():
    hashtags.extend(re.findall(r'#\w+', text))
```

```
hashtag_ctr = Counter(hashtags).most_common(10)
tags, ctr = zip(*hashtag_ctr)
```

```
plt.figure(figsize=(10, 5))
plt.bar(tags, ctr, color='purple')
plt.title(f"Top Hashtags")
plt.xlabel('Hashtag')
plt.ylabel('Frequency')
plt.show()
```

